A/B Test - Subscription in App

Test Goal

The goal of the test is to determine whether the '50% discount' framing increases user conversion to a subscription purchase.

Group A: users are offered a subscription for $4.99.
Group B: users are offered a subscription for $4.99 with a 50% discount.

The key metric is conversion from install to payment.

Data Overview

Metric	Group A	Group B
Users	10,013	9,985
Conversions	611	889
Conversion Rate	6.10%	8.90%
Test Start	2023-07-03	
Test End	2023-07-25	
Test Duration	21 days	

Statistical Testing

To test the null hypothesis (H0: conversion rates in groups A and B are equal), a two-proportion z-test was used.

Z-statistic: -7.519
p-value: 5.49e-14

Conclusion: p-value (5.49e-14) < 0.05, so the null hypothesis is rejected. The conversion rates in groups A and B are statistically different.

Visualization

The chart below compares conversion rates between groups with 95% confidence intervals. Group B shows a clearly higher conversion rate, and the non-overlapping confidence intervals confirm the result is statistically significant.

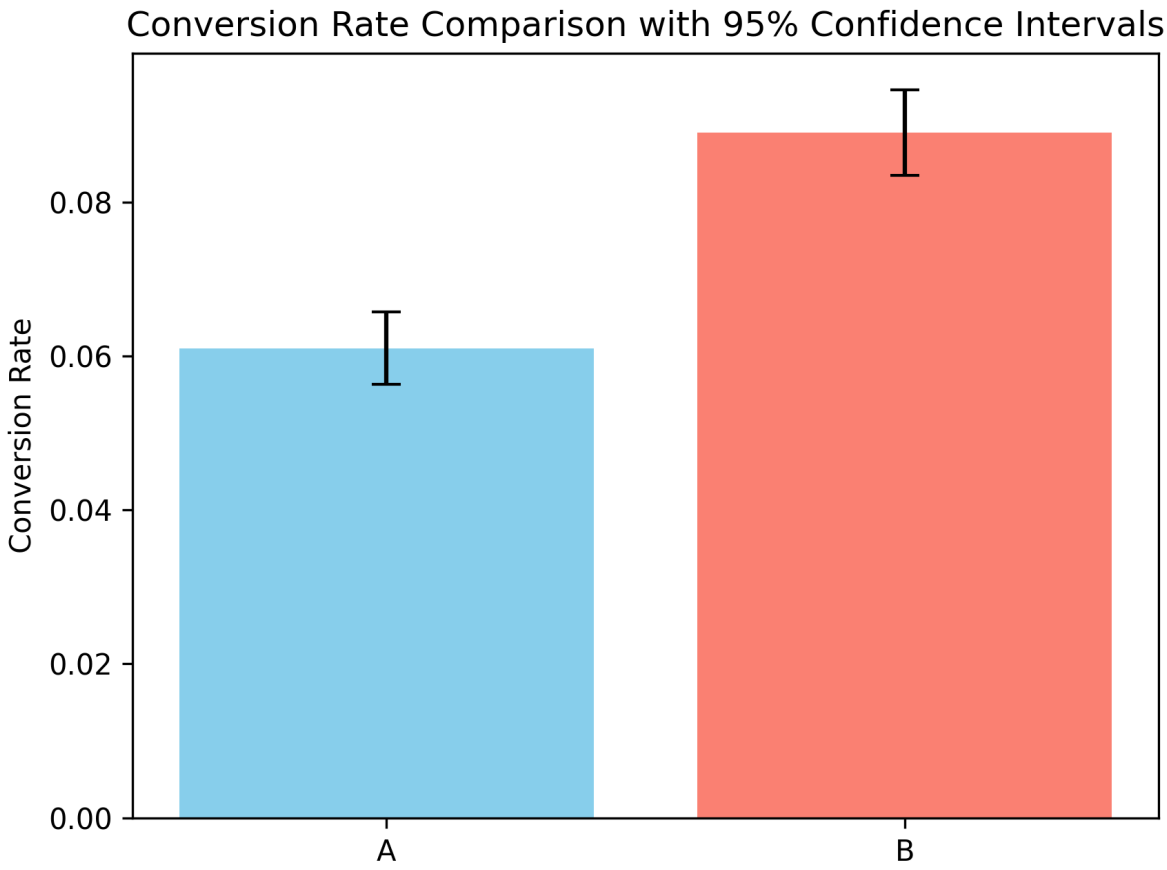

Figure 1. Conversion rate comparison between Group A and Group B with 95% confidence intervals.

The chart below shows the daily conversion rate dynamics over the 21-day test period. Group B consistently outperformed Group A across most days.

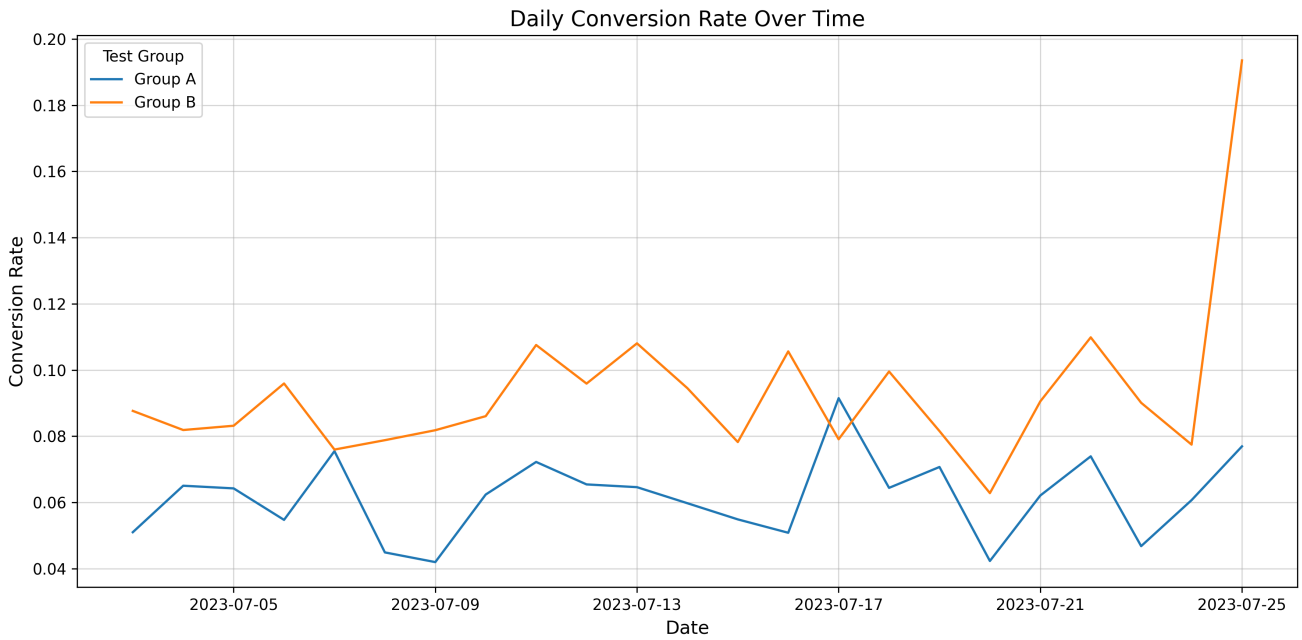

Figure 2. Daily conversion rate dynamics for both groups over the test period.

Conclusions

1. Group B (50% discount framing) showed a higher conversion rate compared to Group A.

2. Based on statistical testing, it can be concluded that the offer framing has a measurable impact on user purchasing behavior.

Recommendations

1. Implement Variant B (50% discount framing) as the primary offer, as the difference is statistically significant ($p < 0.05$) and the effect is not due to random variation.

2. Before full-scale rollout, evaluate long-term user behavior, including retention and subscription cancellation rates.